

Quantifying the value of fertilization advice strategies under weather uncertainty

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Project description

Nitrogen is one of the most important nutrients for crop growth. Too little nitrogen can limit crop development, while excessive fertilization is costly and may have negative environmental effects. Therefore, it is important to apply the right amount of nitrogen at the right time. In practice, fertilizer recommendations are based on measurements of the soil and crop, together with models of crop development, weather forecasts, and other environmental information. However, the quality of a fertilization strategy cannot be judged from the realized crop outcome alone, since this outcome also depends strongly on uncertain future weather conditions. This project aims to quantify the added value of precision fertilization advice by comparing fertilization strategies in terms of expected crop outcomes under weather uncertainty.

CropConsult, a company that provides advice to farmers, has collected field data containing soil measurements, fertilization advice, actual fertilizer applications, measurements of (potato) crop biomass and nitrogen status, weather data, and information about tuber development. These data contain different fertilization decisions made under a range of environmental conditions. The aim is to use this variation to learn how the crop responds to different nitrogen fertilization strategies, and subsequently compare their expected performance under uncertain future growing conditions.

A natural mathematical description in this context is the following:

$$\mathbb{E}[Y \mid \mathbf{N}, \mathbf{x}, \mathbf{U}_{\text{future}} = \mathbf{u}] = f(\mathbf{N}, \mathbf{x}, \mathbf{u}), \quad (1)$$

where Y is the crop yield, $\mathbf{N}$ describes the amount and timing of nitrogen application, $\mathbf{x}$ contains the information available at the time of the fertilization decision, and $\mathbf{U}_{\text{future}}$ represents the uncertain future weather conditions. The function f is unknown and will be estimated from historical field data.

Since future weather is unknown when the fertilization decision is made, the performance of a strategy should not be evaluated for a single realized weather trajectory. Instead, the expected crop yield under the distribution of possible future weather conditions should be considered. This can be expressed as

$$\mathbb{E}[Y \mid \mathbf{N}, \mathbf{x}] = \int f(\mathbf{N}, \mathbf{x}, \mathbf{u}) p(\mathbf{u} \mid \mathbf{N}, \mathbf{x}) d\mathbf{u}. \quad (2)$$

where $p(\mathbf{u} \mid \mathbf{x})$ describes the distribution of possible future weather conditions given the information available at the time of the decision. This makes it possible to compare different fertilization strategies under the same crop and soil conditions while marginalizing over future weather uncertainty. The project will use this framework to quantify how much precision fertilization advice improves expected crop yield, and under which conditions this added value is largest.

Reference for colloquium

T. Gneiting and M. Katzfuss, “Probabilistic forecasting,” *Annual Review of Statistics and Its Application*, 1, 125–151, 2014.

Other references

B. Dumont, B. Basso, B. Bodson, J.-P. Destain, and M.-F. Destain, “Climatic risk assessment to improve nitrogen fertilisation recommendations: A strategic crop model-based approach,” *European Journal of Agronomy*, 65, 10–17, 2015.